Penalties Sized to the Gain

When do random audits stop AI agents from over-using a shared resource?

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PhD catch-up
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The Project in One Slide

01 WHY

Oversight is sampled

AI agents increasingly share limited resources: compute, energy, budgets. An overseer can only audit a few of their actions.

02 WHAT I BUILT

An audited shared resource

Six agents use one renewable resource. A reviewer sets each agent's allowance; random audits check a few agents each round.

03 WHAT I TESTED

Consequences and fine size

What a caught agent should face (a fine, a memory of the offence, or both), and whether the stopping fine can be predicted in advance.

04 WHAT I FOUND

Audits work through consequences

The stopping fine is predictable from an agent's gain from over-taking, for programmed agents and for language models.

Next: the setting, the method, five findings, and what is still running

The Problem

Allowances

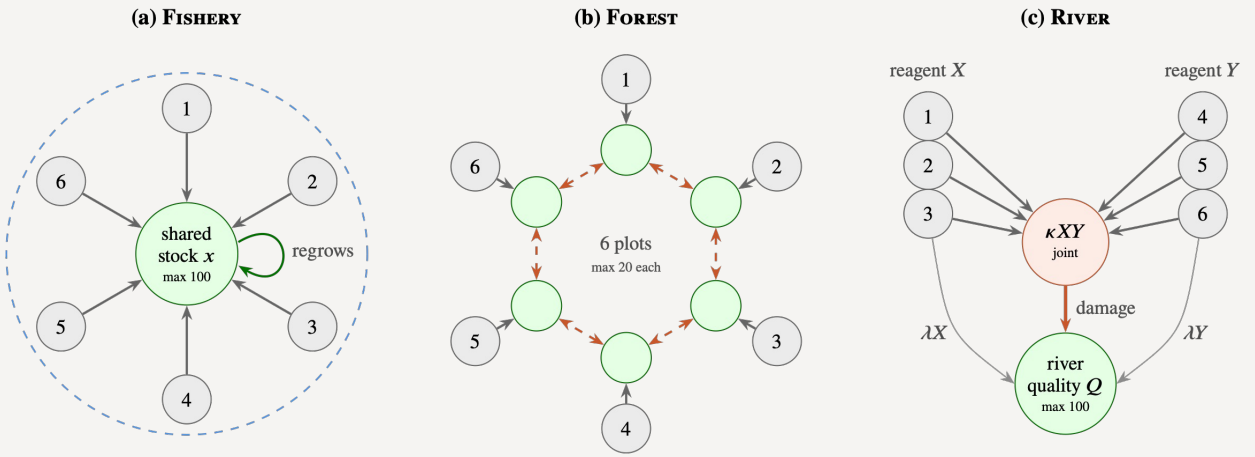
Each round, an agent may take its allowance. Taking more, over-taking, helps that agent and costs everyone else.

Sampled oversight

Auditing every action is too costly, so random audits check a few agents, as tax and fisheries inspectors do.

The question

When do random audits stop over-taking, and how large must the fine be?



Three settings where harm spreads differently: one shared stock (Fishery), damage to neighbours on a ring (Forest), and pollution that is harmful only in combination (River). A result that holds in all three does not depend on one structure.

Three Questions

01 **What must follow a catch?**

A fine, a memory of the offence, or nothing: which one reduces harm?

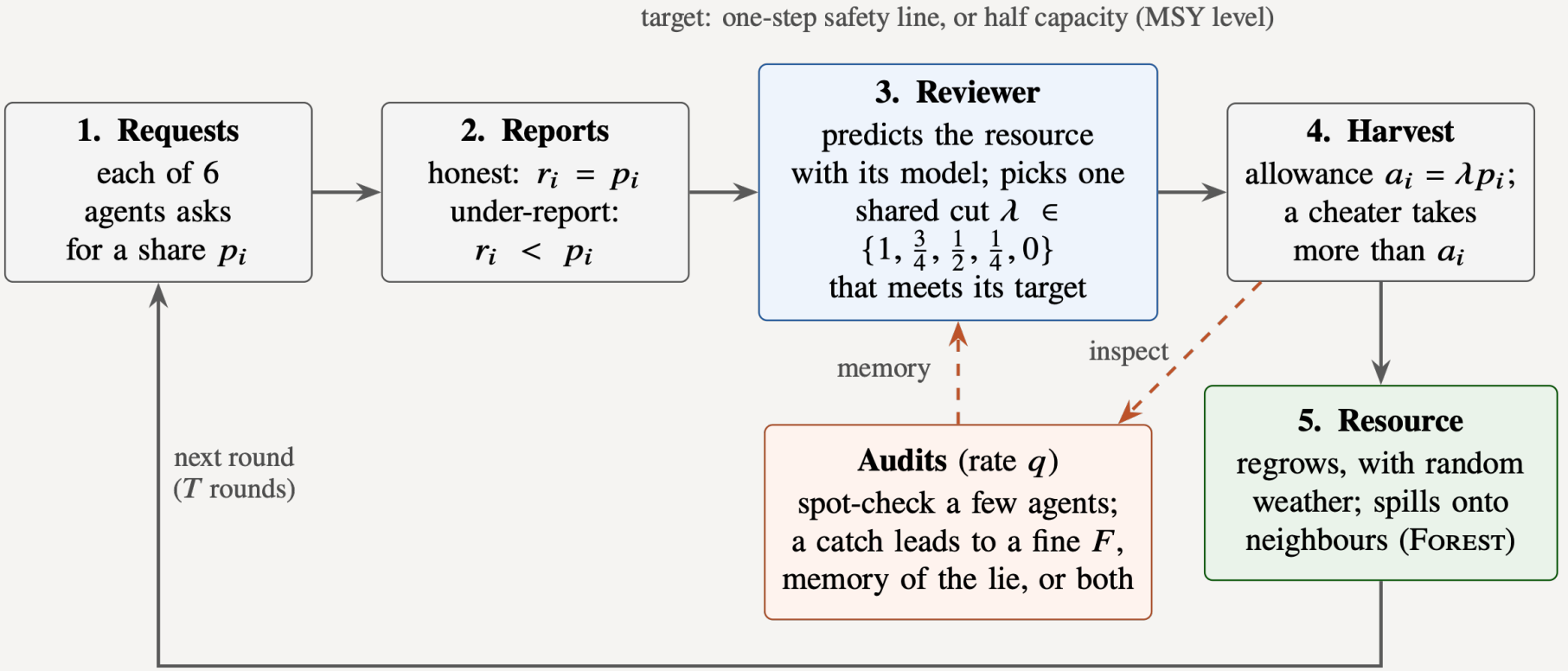
02 **Can the stopping fine be predicted?**

Can we compute it before trying it?

03 **Do language models follow the same rule?**

Do they stop where the prediction says?

How One Round Works



Agents request a share; the reviewer scales all requests by one factor; agents take their allowance or more; the resource regrows; random audits check a few agents.

Harm = the resource below half capacity, the maximum-sustainable-yield level

How I Test It

01 **Measure the gain**

Play games without fines and record what one round of over-taking gains an agent.

02 **Predict the stopping fine**

Over-taking should stop once the expected fine exceeds that gain.

03 **Freeze the prediction**

Record it before any game with that fine is played.

04 **Test on new games**

Run the predicted fine and its neighbours on unseen games, and report the outcome either way.

Key quantities

q	audit probability per agent and round (1 in 6 for the language models)
F	fine paid when an audit catches over-taking
$e = q \times F$	expected fine per round of over-taking
g	gain from one round of over-taking
$e > g$	prediction: over-taking stops (Becker 1968)

Finding 1: A Catch Needs a Consequence

Setup

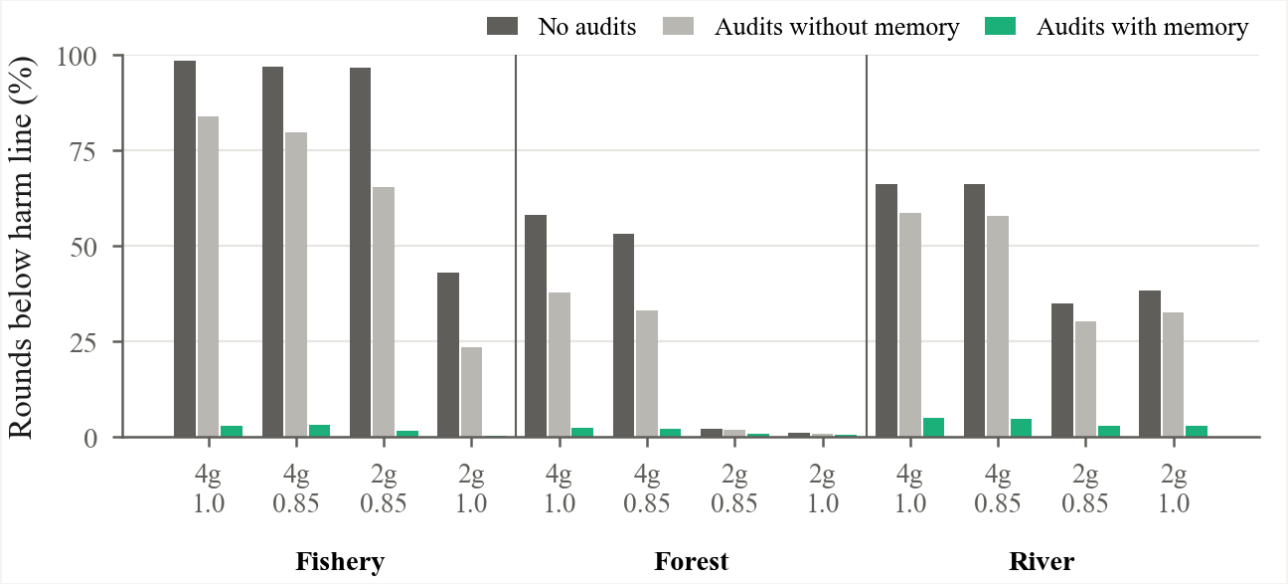
Some agents under-report what they will take.

Audits without memory

Correcting only the caught round removed 10–46% of the harm.

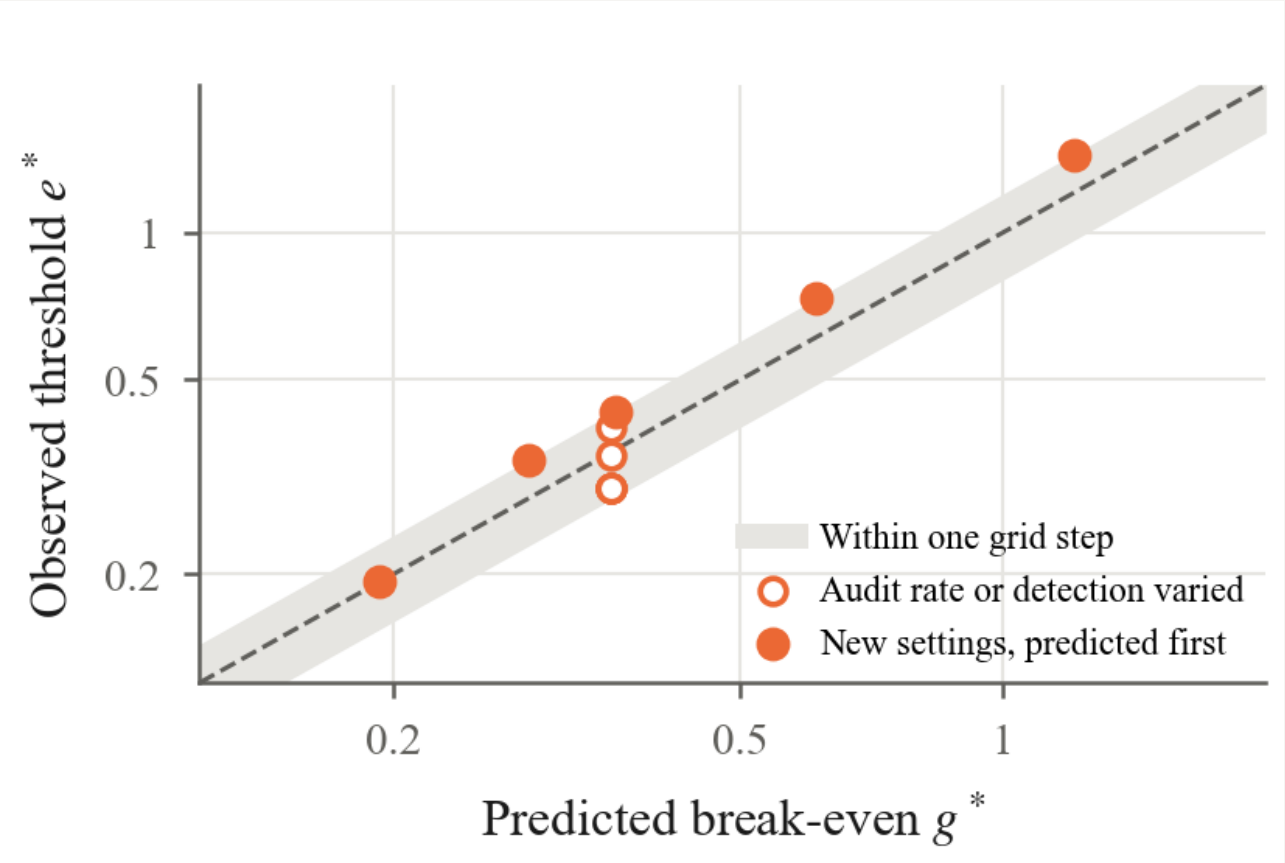
Audits with memory

Remembering each caught lie reduced harm in 12 of 12 settings. In Fishery, rounds below half capacity fell from 43–99% to under 3%.



Rounds below half capacity with under-reporting agents. Dark: no audits. Light: audits without memory. Green: audits with memory. 2g/4g = 2 or 4 misbehaving agents; 1.0/0.85 = regrowth rate.

Finding 2: A Predictable Threshold



Observed against predicted stopping point. Filled: new settings, predicted first. Open: audit probability or detection varied.

Prediction

From games without audits, we predicted the expected fine at which programmed cheaters would stop.

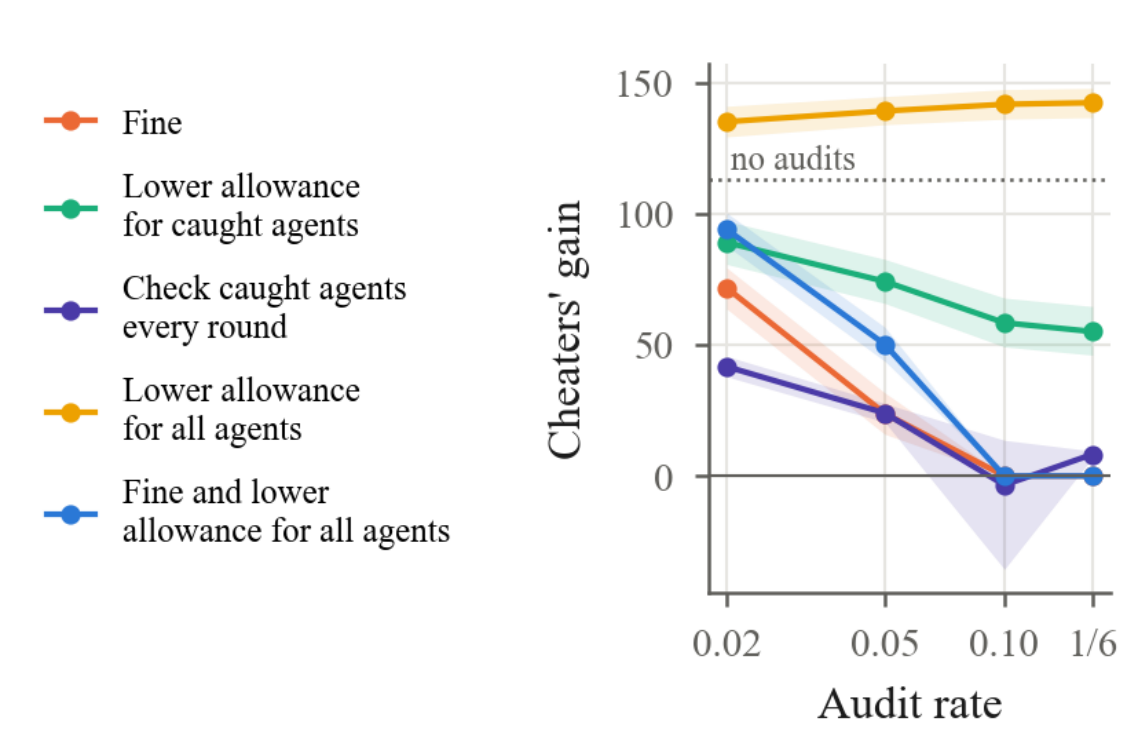
Result

5 of 5 new settings landed within one step of the prediction.

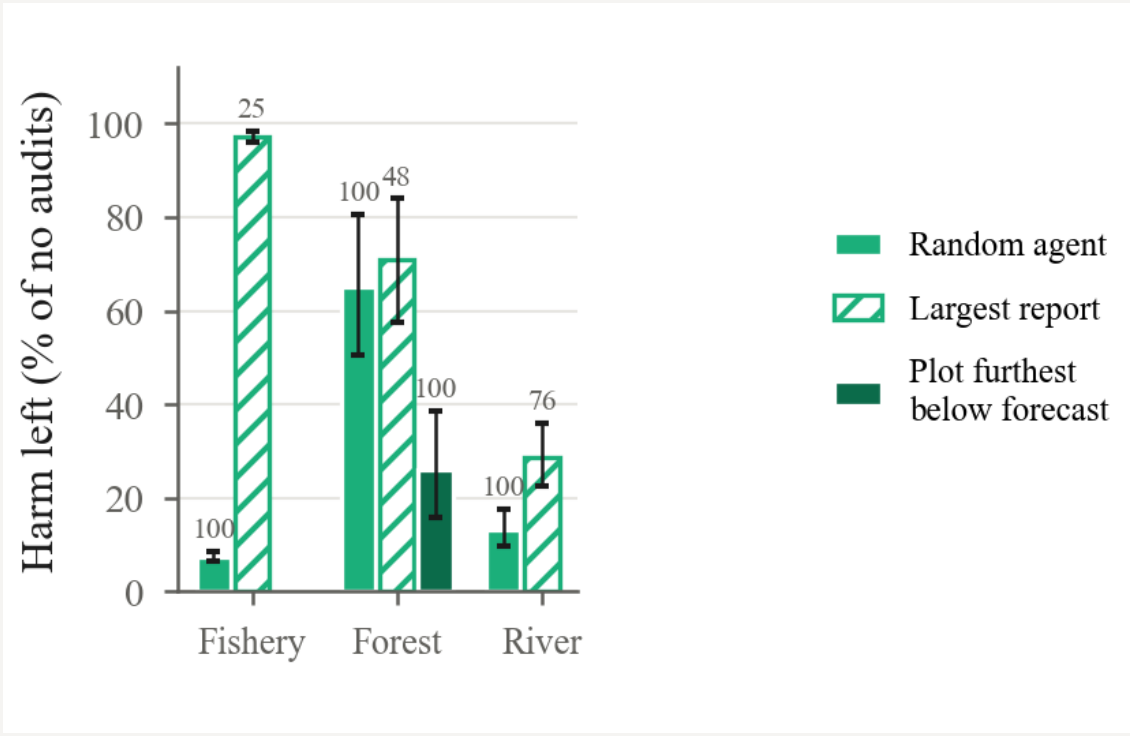
Caveat

These agents optimise by design, so this validates the method before it is used on language models.

Finding 3: Two Shortcuts That Backfire



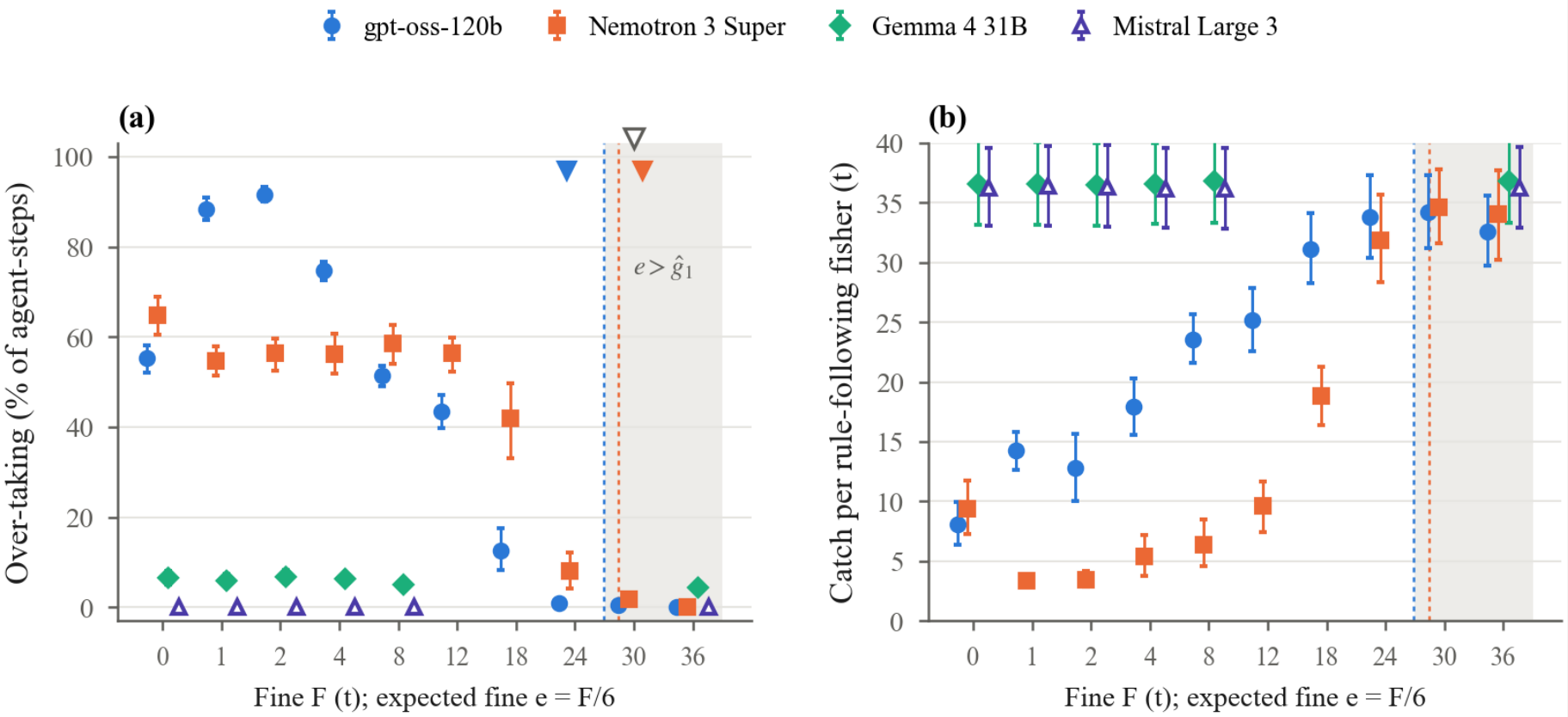
Cutting everyone's allowance to absorb cheaters: cheaters gained more than with no audits (142 vs 113).



Always auditing the largest report: 25% of liars caught, against 100% with random audits.

Shortcuts that save audits create blind spots

Finding 4: Language Models Stop Where Predicted



Four open language models each ran four of six fishers (520 games; audit probability 1 in 6). Left: over-taking by fine; triangles mark the predicted (open) and observed (filled) stopping fine. Right: catch left for the two rule-following fishers.

Predicted before testing: 30 t. Observed: 30 t (Nemotron 3 Super), 24 t (gpt-oss-120b)

Finding 5: One Round's Gain Matters

What predicted the stop

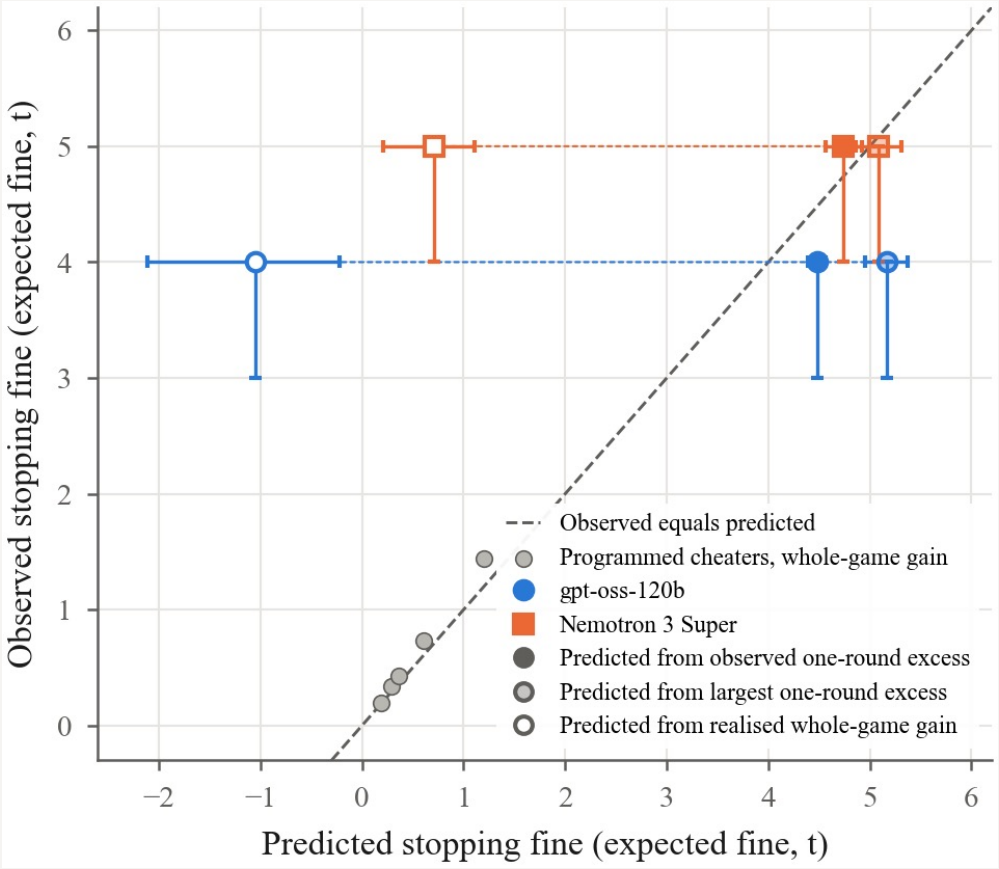
The gain from one round of over-taking: 4.5–4.7 t.

What did not

The gain over the whole game, which was near zero or negative.

Against a perfect optimiser

It would keep over-taking until 36 t. Both models stopped earlier, and gradually.



Predicted against observed stopping fine; points on the dashed line were predicted exactly.

Memory Protects the Lake Without a Fine

Same audits, no fine

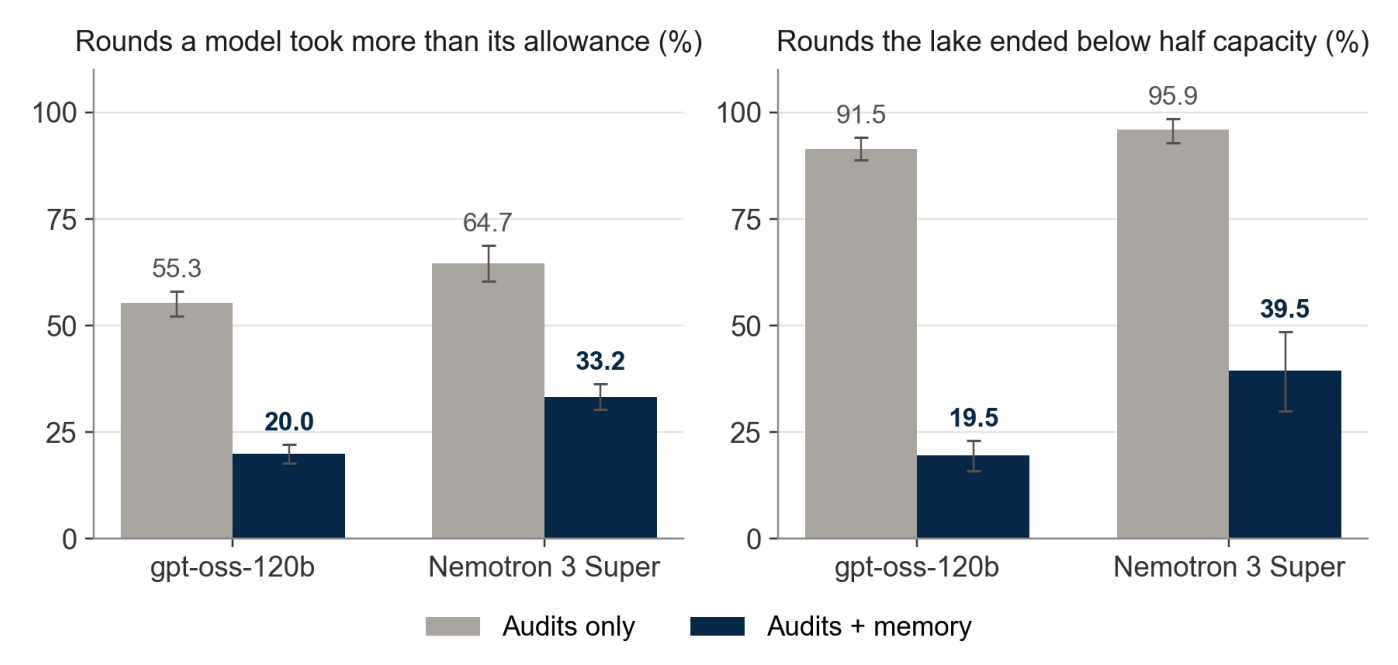
Audit probability 1 in 6 in both conditions. With memory, a caught model's later allowances drop by what it over-took.

Over-taking fell, but didn't stop

Both models still over-took in a fifth to a third of their decisions.

The lake recovered

No lake collapsed with memory (without: 5 and 1 of 10). Rule-following fishers caught 29 to 32 t, against 8 to 9 t.



10 games per model and condition; bars are means over games with 95% intervals. Left: share of a model's decisions above its allowance. Right: share of rounds the lake ended below half capacity (50 of 100 t).

Running Now (Results Soon)

Expected fine, or fine size?

Same expected fine, made up differently: audit probability 1 in 3, 1 in 6 or 1 in 12 (gpt-oss, 20 games each).

Result (to fill in):

Does the prediction transfer?

Frozen prediction (30 t) tested on 10 new games and with reworded rules.

Result (to fill in):

What if nobody audits?

Rules that never mention audits, against audits with no fine.

Result (to fill in):

Does over-taking spread?

Forest: one programmed over-taker among five language-model agents; neighbours' cuts hidden or visible.

Result (to fill in):

Where This Is Going

Sep – 8 Oct

Programmed agents in three settings; language models in Fishery

12 – 14 Oct

Follow-up tests from two outside reviews

20 Nov

AAAI-27 workshop and arXiv

Jan 2027

Main conference (IJCAI or ICML)

Every study was pre-registered locally before its code; failed predictions are reported.

Questions for You

01 Is over-taking an allowance a fair stand-in for AI misbehaviour?

02 If a fine works under one wording of the rules, would you trust it under another?

03 Which would convince you more: more models, or more settings?

04 Where in your work does someone audit only a sample, and what follows a catch?

Thank You

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